

TYUMEN ROSCHINO, Russian Federation

WMO: 232204

Lat: 57.190N

Lon: 65.324E

Elev: 115

StdP: 99.95

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.0	(c) -29.2	(d) -35.1	(e) 0.1	(f) -31.7	(g) -32.2	(h) 0.2	(i) -28.9	(j) 9.1	(k) -8.1	(l) 8.4	(m) -9.1	(n) 2.6	(o) 230	(p) 0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 30.0	(d) 19.9	(e) 28.1	(f) 19.0	(g) 26.2	(h) 18.1	(i) 21.2	(j) 27.1	(k) 20.3	(l) 25.9	(m) 19.4	(n) 24.8	(o) 4.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.2	(b) 14.1	(c) 23.7	(d) 18.2	(e) 13.3	(f) 22.5	(g) 17.2	(h) 12.5	(i) 21.6	(j) 62.2	(k) 27.1	(l) 59.1	(m) 25.9	(n) 56.1	(o) 24.6	(p) 24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.2	7.4	-34.6	32.8	3.5	1.5	-37.1	33.8	-39.1	34.7	-41.1	35.5	-43.6	36.6
			-34.7	22.5	3.3	0.9	-37.1	23.1	-39.0	23.7	-40.8	24.2	-43.2	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-16.5	-13.6	-4.7	4.8	11.7	17.1	18.8	16.6	10.6	3.2	-5.6	-12.6		
	DBStd	13.50	7.47	6.79	4.86	5.30	5.32	4.57	3.85	4.18	4.30	5.16	7.20	7.73		
	HDD10.0	3586	822	660	454	169	45	4	0	2	43	218	468	700		
	HDD18.3	5892	1080	893	713	406	215	76	43	86	234	471	718	958		
	CDD10.0	875	0	0	0	14	97	218	274	205	62	6	0	0		
	CDD18.3	139	0	0	0	0	8	39	58	31	2	0	0	0		
	CDH23.3	1306	0	0	0	4	115	389	509	269	19	0	0	0		
	CDH26.7	338	0	0	0	0	21	112	128	75	2	0	0	0		
(14) Wind	WSAvg	4.0	3.9	3.8	4.5	4.5	4.3	3.9	3.7	3.7	3.8	4.2	4.2	4.0		
(15) Precipitation	PrecAvg	485	22	15	23	24	42	59	88	62	49	39	35	27		
	PrecMax	672	47	33	55	78	104	126	154	114	113	93	70	53		
	PrecMin	352	2	2	2	1	5	18	15	15	10	2	2	6		
	PrecStd	82	12	8	13	20	25	29	37	24	26	24	16	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	3.1	8.8	23.0	29.1	32.0	31.9	31.2	25.9	18.9	9.0	3.0		
		MCWB	0.8	1.1	4.3	13.3	16.2	19.3	20.6	19.9	15.8	11.8	6.1	1.8		
	2%	DB	-1.2	1.0	5.8	18.9	26.2	29.8	29.8	28.9	22.2	15.2	6.0	1.1		
		MCWB	-1.6	-0.3	2.3	11.0	14.8	19.2	20.7	19.7	15.1	9.8	5.0	0.4		
	5%	DB	-3.8	-1.8	3.9	15.8	24.1	27.8	28.0	26.2	19.9	12.8	4.2	-0.2		
		MCWB	-4.2	-2.9	1.3	9.0	14.0	18.6	19.7	19.1	14.3	8.9	3.5	-0.6		
	10%	DB	-6.0	-4.0	2.1	13.0	21.2	25.2	26.2	24.0	17.2	10.1	2.8	-2.8		
		MCWB	-6.5	-5.0	0.3	7.6	12.8	17.9	18.7	18.0	12.8	7.7	2.1	-3.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.4	4.5	14.0	18.5	22.1	22.6	21.6	17.6	12.4	6.5	1.7		
		MCDB	1.1	2.6	7.8	19.4	24.8	27.4	28.2	27.6	23.3	17.4	7.9	2.3		
	2%	WB	-2.0	-0.4	2.7	11.6	16.6	20.6	21.5	20.6	16.1	10.7	5.0	0.4		
		MCDB	-1.6	1.0	4.8	17.8	23.1	26.6	27.2	26.6	20.7	14.5	5.7	1.0		
	5%	WB	-4.1	-2.9	1.6	9.8	15.1	19.5	20.7	19.6	14.8	9.2	3.6	-0.9		
		MCDB	-3.7	-1.8	3.1	14.6	21.9	25.4	26.2	25.0	18.6	11.9	4.2	-0.5		
	10%	WB	-6.5	-5.0	0.5	7.9	13.5	18.5	19.7	18.6	13.4	7.9	1.8	-3.2		
		MCDB	-6.2	-4.0	2.0	11.9	20.1	24.0	24.8	23.3	16.9	10.2	2.4	-2.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.9	8.8	9.4	11.6	10.7	9.9	9.7	8.7	7.0	5.7	6.4		
		MCDBR	6.4	7.9	9.1	13.2	14.6	13.6	12.7	12.8	11.7	10.7	6.0	5.3		
	5% WB	MCWBR	6.2	7.0	6.6	7.2	6.9	5.9	5.4	5.7	6.1	6.4	5.1	5.1		
		MCWBR	6.3	8.0	8.0	12.0	13.0	11.9	11.3	11.7	10.6	9.8	5.9	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.309	0.366	0.407	0.404	0.403	0.419	0.427	0.380	0.346	0.303	0.278		
		taud	2.040	2.038	1.957	2.098	2.240	2.314	2.272	2.218	2.314	2.271	2.132	1.998		
	Ebn at Noon		554	706	756	796	831	838	812	773	757	672	524	429		
		Edh at Noon	60	93	132	135	126	118	121	120	94	74	55	47		
(44) All-Sky Solar Radiation	RadAvg		0.70	1.69	3.19	4.44	5.20	5.70	5.47	4.20	2.64	1.34	0.65	0.42		
	RadStd		0.05	0.12	0.18	0.35	0.44	0.44	0.49	0.41	0.27	0.21	0.09	0.05		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page